Build a route with arrow cards

Name: ____________________ Team: ____________________ Date: __________

An L-shaped route made from short drives and a turn.

Gather these materials

[ ] Checked base, ruler and removable floor tape

[ ] Six paper cards about 8 cm square

[ ] Pencil or marker

[ ] The short-drive code from lesson 2

Build: steps 1-4

[ ] 1. Draw forward arrows on four cards, a right-turn arrow on one card and STOP on the last card. These cards are a plan; they do not control the robot by themselves.

[ ] 2. Mark a short straight lane and a wide delivery space on the floor. Leave enough clear room for a turn. Keep arrow cards beside the lane, away from the wheels.

[ ] 3. Arrange the cards: drive forward, turn right, drive forward, stop. Point to the direction the robot will face after each step.

[ ] 4. Test the first short drive on its own. Adjust its time until it reaches the turning area. Calibrate means test and adjust a setting.

Before power: parts secure, wheels clear, Stop controls easy to reach.

Before a run: clear the lane and ask your teacher to check the setup.

After a run: stop and turn off the robot before changing any parts.

Drawings show the idea. Measure your own pieces; drawings are not full size.

Finish, code and test

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Build: steps 5-8

[ ] 5. With the teacher, test a short, slow right turn by itself. Stop after the turn. Change its time a little if it turns too far or not far enough.

[ ] 6. Test the second short drive. Join the three checked parts, with a Stop after each one. Keep the first complete tests to 3 seconds of movement or less.

[ ] 7. Run the route three times from the same mark and direction. Record where the robot stops each time. Keep people and objects out of the lane.

[ ] 8. Change one time setting and try another set of runs. Save the code and draw the final route.

Make the program

Start stopped. Drive slowly for the tested first time, then stop.

Turn right slowly for the tested turn time, then stop.

Drive slowly for the tested second time, then stop and end.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

Record your test runs

Run	First / turn / last times	Inside space?	What to adjust?
1			
2			
3			

A successful build

The program stops after each part and at the end.

The robot finishes inside the space in at least two of three runs.

The team records the times and layout so it can try again.

Which time setting did you change, and what happened?